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A comparative study of the effects of urban morphology on land surface temperature in Chengdu and Chongqing, China

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Urbanization combined with global climate change, exacerbates the urban thermal environment and hinders sustainable urban development. However, the complex relationships between land surface temperature (LST) and urban morphology are being further understood, particularly in relation to different urban development patterns, distinct topography, and 3D building morphology. Thus, this study conducted a comparative study in Chengdu and Chongqing, Southwest China. We explored the impact of comprehensive factors (including socio-economic factos, topography, land use composition, and building morphology) on LST by employing the methods of linear regression, geographical detector model, and the boosted regression trees. Our results suggest that (1) high LST was mainly observed in the central part of Chengdu but it presented multicenter aggregation trend in Chongqing; (2) Socio-economic factors were the dominant variables affecting LST in both cities; (3) land use composition and building morphology showed distinct contributions to LST among the two cities; and (4) 3D building management was more effective in Chengdu than in Chongqing. A better understanding of the impact of various influencing factors on LST will enable policy makers and planners to develop appropriate strategies for constructing climate-adaptive cities and mitigating urban heat.

Keywords Urban morphology, Land surface temperature, 3D buildings, Boosted regression trees, Comparative study

Accelerated urbanization occurs worldwide, even forming urban megaregions and gigacities somewhere[1–3]. Previous studies pointed out that the area of urban land increased at an average rate of 9.7×10^3 km^2/year from 1985 to 2015[4]. It is projected that the global urban population will exceed 68% of the total population in 2050[5]. Rapid urbanization processes were accompanied by abundant farmland/natural land being replaced by impervious surfaces, which significantly changed the near-surface ecological processes[6,7]. Ultimately, these changes have led to a series of ecological challenges, such as urban heat islands (UHIs), great losses of ecosystem services, and urban flooding[8–10]. UHI, arguably the most prominent urban climate phenomenon, refers to the higher temperature in urban space than its surrounding areas. UHIs can potentially cause to a increase in energy consumption, deteriorate air quality, and negatively impact the health of urban residents[11–13]. Heatwaves, another type of extreme weather, frequently occur. The death toll from European heatwaves reached over 1000 in 2022. UHIs and frequent heatwaves deteriorate the urban thermal environment, which severely challenges the sustainable development of cities. Thus, the mechanisms of heat mitigation and adaptation have become urgent and crucial[14–18].

Air temperature and land surface temperature (LST) are commonly used to measure urban thermal conditions[19,20]. Air temperature is commonly collected from meteorological observations. The meteorological observation method has the advantages of high temporal resolution and long-term collection, but limited by the number of observation stations, it has the problem of insufficient spatial heterogeneity[21]. Compared to the conventional meteorological observation method, the remotely sensed LST is better suited to explore urban thermal conditions at both large-scale and intraurban scales[22,23]. Numerous studies have identified the spatial-temporal patterns of LST across various scales[24–26]. The LST, determined by incoming solar radiation, energy allocation, and anthropogenic heat, is impacted by urban morphology and social-economic factors[27–31]. Among

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these influencing factors, urban morphology, which includes land cover composition and configuration in horizontal direction and buildings-vegetation composition and configuration in vertical direction, exerts a more significant influence than the other factors in a particular city. Specifically, 2D urban morphology mainly refers to the spatial composition and layout of the city on a two-dimensional plane, and 3D urban morphology mainly refers to spatial composition and layout of the city in three dimensions. Monocentric/polycentric forms represents the patterns of commercial land in horizontal and vertical direction, commercial districts are commonly the highest LST zone in the city.

Comprehensive analysis/exclusive investigation of the contributions of multiple influencing factors to LST have been significantly conducted[32,33]. For example, Logan et al. (2020) demonstrated the influence of urban characteristics (urban morphology and social-economic factors) on LST across four cities in the United States[31]. Sun et al. (2020) comprehensively explored the impact of 3D building morphology on seasonal LST in neighborhoods in Shanghai, China[22]. Numerous studies have studied the relationships between urban morphology and LST[34–38]. However, there remains some debates regarding several key points: (1) What is the role of topography in affecting the LST? (2) How urban morphology affects LST in unique geographical environments?

Benefiting from the development of three-dimensional data availability and the 3D landscape index, comprehensive 3D urban morphological factors were taken into account in the urban climate field[22]. A pioneering work was the development of local climate zones (LCZs). This frame determines the LCZs by integrating the land cover and building morphology[15]. But whether 3D urban morphology plays a dominant role in regulating LST is controversial. Some studies have suggested that 3D factors are not important[14,31]. But other studies found that 3D urban characteristics significantly affect LST[39–41]. Additionally, 2D/3D urban morphology plays an equally important role in regulating air temperature in neighborhoods in Beijing[42]. Beyond the 2D or 3D urban morphology debate, there are other important influencing factors, such as green space and socio-economic factors, which could also affect urban LST[43]. Thus, a comprehensive understanding of the importance of 2D/3D urban morphology remains insufficient.

Fewer studies have concentrated on the impact of monocentric/polycentric urban development forms on LST[44]. The monocentric form refers to a large urban center that dominates every level, even eliminating the existence of intermediate layers. The polycentric form refers to the existence of multiple urban agglomerations with similar sizes at different levels of the urban hierarchy. There are no certain conclusions that polycentric urban spatial morphology has a better performance in decreasing LST than the monocentric form[27,45]. An investigation found that the polycentric form was beneficial for heat mitigation in China[44,46]. The results from a comparative study indicated that the polycentric form has a lower UHI intensity than the monocentric form[47]. Some studies have inferred that the polycentric form may deteriorate the urban thermal environment[48]. Additionally, other scholars believe that the polycentric form has a limited influence on the SUHI[45,49]. Thus, it is necessary to further investigate the relationships between LST and monocentric/polycentric urban forms.

Relatively few studies have considered the influence of topography on LST. Because most existing studies have focused on plain cities, which overlooked the slight variations of topography within the cities[33,50,51]. These studies assumed that minor changes in topography have little impact on UHI intensity[52]. However, Even in plain areas (with small variations in elevation), elevation still shows a positive correlation with LST[41]. Considering the increase in urban land expansion worldwide and the potential risk of high temperatures, it is important to further investigate the role of topography in regulating LST[53]. Existing studies have shown that topography can have an important impact on LST[54]. For example, the mountain coverage ratio was significantly negatively correlated with LST in the mountain city of Guiyang, China[55]. Therefore, topography has different effects on the LST across plain cities and mountain cities. Specifically, topography exhibited a weak effect on LST in the plain city, whereas it exerted a strong effect in the mountain city[47]. In this sense, further investigation is needed to understand the different effects of topography on LST across different types of cities.

Most studies have only selected partial influencing factors and adopted a single method for analysis, which limits the universality of the conclusion. Besides, many studies adopted correlation analysis and general linear regression due to their simplicity and clarity, but these methods have limitations in explaining the interdependencies between influencing factors[43,56]. Also, these methods are sensitive to multicollinearity. Furthermore, these methods fail to quantify the relative contributions of the influencing factors and do not consider the spatial heterogeneity of LST. In contrast, the methods that take spatial heterogeneity into account have got increasing attention, such as GWR and GD. These methods quantify the contribution of individual factors, improve their explanatory power for LST, and capture the interactions between factors. Additionally, machine learning algorithms could easily map the marginal effects of the selected indicators and are gradually gaining increasing applications in the urban heat island field[22,31,41,57]. At present, comprehensive consideration of these factors, quantification of their relative importance, and mapping of their marginal effects are still lacking.

The Chengdu-Chongqing Urban Agglomeration is an important platform for the Western Development, a strategic support for the Yangtze River Economic Belt, and an important demonstration area for the country to promote new urbanization. With the release of "The Belt and Road Initiative" and "Yangtze River Economic Belt Strategy", the Chengdu-Chongqing Urban Agglomeration will form an important growth pole for high-quality development. It is crucial to balance ecological protection and economic development. Thus, taking the Chengdu-Chongqing Urban Agglomeration as a case study is meaningful in heat island mitigation and reduction for inland cities. Chengdu and Chongqing's central cities were selected as the research areas for several reasons. First, Chengdu and Chongqing have similar climate zones and suffer UHI effects[58,59]. Second, their urban development patterns differ, as Chengdu is a monocentric city, while Chongqing is a polycentric city. Third, they have distinct landform, with Chengdu located on an alluvial plain and Chongqing located in a mountainous region. This study highlighted that different landforms had distinct influences on urban

morphology, particularly the monocentric/polycentric forms. According to previous studies, the differences in monocentric/polycentric urban form induced the UHI differences.

In summary, the comprehensive influencing factors of LST need to be explored in terms of monocentric/polycentric urban form, topography, 2D/3D, interactive effects, and marginal effects. Here, we aimed to address two issues: (1) What are the dominant influencing factors and their marginal effects? (2) How do the relationships vary across two cities with distinct landform and urban form? To answer these questions, we conducted a comparative study to investigate the contributions of urban morphology indicators and the interactive effects by employing linear statistics, machine learning and Geo-Detector techniques. Further understanding the relationships between LST and urban morphology can provide appropriate strategies for urban heat adaptation and mitigation.

Methods

Study area

The Chengdu-Chongqing Urban Agglomeration, which is located in the upper reaches of the Yangtze River, covers an area of 1.8×10^5 km^2 (Fig. 1). Chengdu and Chongqing are in a leading position in the Chengdu-Chongqing Urban Agglomeration and are commonly referred to as "twin cities". Chengdu and Chongqing both have a subtropical monsoon humid climate. The Chengdu-Chongqing Urban Agglomeration has witnessed rapid urbanization, especially since the implementation of the Western Development Policy in 1999. During 2000–2020, urban areas increased by 4393 km^2. In 2020, the urban populations of Chengdu and Chongqing reached 16.49 million and 22.26 million, respectively. The Belt and Road Initiative was announced in 2013, which promoted inland urban prosperity.

Data sources

The population density data in 2021 were extracted from the world population density map published by Worldpop (https://www.worldpop.org/) (Table 1). The Point of Interest (POI) was collected from the Gaode Map (https://www.amap.com/), which includes food services, medical facilities, shopping areas, leisure and entertainment, attractions, transportation, public facilities, and accommodations. The POI density was calculated using kernel density analysis in ArcGIS 10.8. The nighttime light data in 2020 were obtained from the 2012–2020 Global Annual Nighttime Light Dataset (Annual VNL V2) (https://www.ngdc.noaa.gov/eog/). Digital elevation model (DEM) data (resolution: 30 m) were obtained from the Geospatial Data Cloud (http://www.giscloud.cn/). A digital elevation model was used to calculate urban surface roughness (USR). Land use data in 2020 (resolution: 10 m) were collected from the European Space Agency (ESA) (https://viewer.esa-worldcover.org/worldcover),

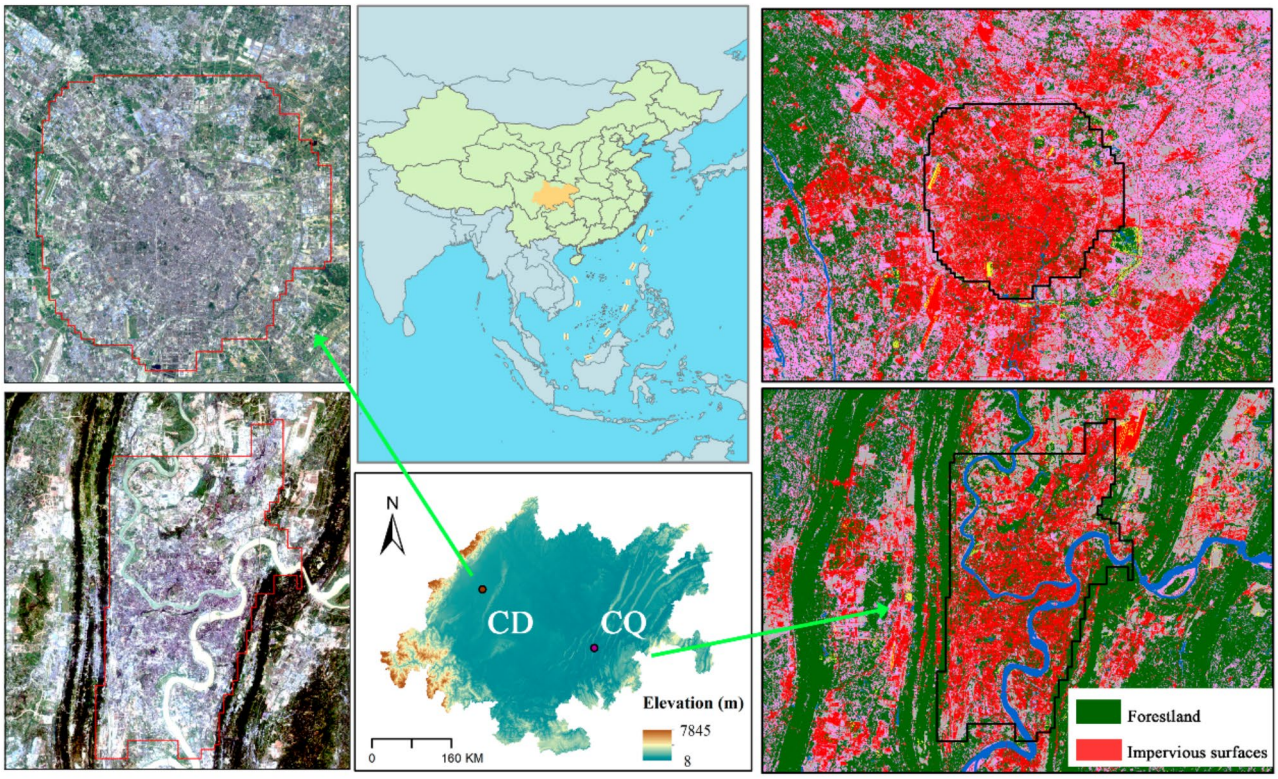


Fig. 1. Location of the study area.The maps were created using ArcGIS, version 10.7, (https://www.esri.com/en-us/arcgis/about-arcgis/overview) and Adobe Photoshop, version 13.0, (https://www.adobe.com/cn/creativecloud/roc/business.html).

Category	Name	Abbr.	Description	Formulas
Socio-economic factors	Population density	POP	Population per unit land area.	$POP = \frac{P}{Ssite}$
	POI density	POI	Total number of POI per unit land area.	$POI = \frac{N}{Ssite}$
	Night light intensity	NL	Light intensity from light source in night environment.	$NL = \sum_{i=1}^{n} DN$
Terrain	Urban surface roughness	USR	The relative elevation difference of the grid.	$USR = E_{max} - E_{miin}$
Building morphology factors	Building coverage ratio	BCR	Building coverage ratio of the grid.	$BCR = \frac{\sum_{i=1}^{n} BSi}{Ssite}$
	Building height	BH	Mean building height of the grid.	$BH = \frac{\sum_{i=1}^{n} BH_i}{n}$
	3D shape index	3DSI	The degree of shape complexity.	$3DSI = \frac{\sum_{i=1}^{n}(BSi + Li \times BHi)}{3 \times v \times \sqrt{3v/4\pi}}$
	3D fractal	3DF	The compactness of the buildings.	$\lg N(r) = -\mathrm{D}\lg r + c$
Land cover composition	percentage of impervious surface	PIS	The ratio of impervious surface area to grid area.	$PIS = \frac{\sum_{i=1}^{n} A_i}{Ssite}$
	percentage of green space	PGS	The ratio of green space to grid area.	$PGS = \frac{\sum_{i=1}^{n} A_{gs}}{Ssite}$

Table 1. Description of the influencing factors. Notes: P: population size; N: total number of POIs; DN : light value per pixel; E_{max}: the max elevation of the grid; E_{min}: the minimum elevation of the grid; BS_i: building projected area; n: the number of buildings; S_{site}: area of the grid. BH: mean building height; BH_i the building height; F: floor of building. c: constant. Li: perimeter of the buildings. v: volume of buildings of the grid. $N(r)$: number of noneempty boxes; r: scale; D: 3D fractal dimension. $Dbuilding$: mean distance between buildings and their nearest neighbor. Si is the overlapping area of buildings; A_i is the area of impervious surface area; and A_{gs} is the area of green space.

with an overall accuracy exceeding 85%. The percentage of impervious surface (PIS) and percentage of green space (PGS) were extracted and calculated by adopting Fragstats 4.2. Building data in 2020 were obtained from Baidu Maps (https://map.baidu.com/), which includes location, footprints, and floor information. By referring to the high-resolution image in 2020 in Google Earth Pro software, we found that the location is highly accurate. The height of the buildings, assuming a per floor height of 3 m, was validated by Baidu Quanjing Maps. Four building morphology indicators were selected, including the building coverage ratio (BCR), three-dimensional fractal dimension (3DF), building height (BH), and three-dimensional shape index (3DSI).

Specifically, the population density data used in this study has a high consistency with the population density data in 2020 collected from The Resources and Environmental Scientific Data Center (RESDC), namely Chinese Kilometer Grid Population Distribution Dataset.As you know, the Chinese Kilometer Grid Population Distribution Dataset comprehensively considers multiple factors closely related to population, such as land use types, nighttime light brightness, and residential density, and uses a multi factor weight allocation method to distribute population data based on administrative regions onto a spatial grid, thereby achieving spatialization of population. So, the Chinese Kilometer Grid Population Distribution Dataset is reliable and the data used in this study is reliable too.

Methods

Analysis unit

In terms of a single city, the most common analysis units were block, neighborhood, and grid scales. Compared to the block/neighborhood units, the grid scale was beneficial in indicating the comprehensive attributes, because the city is a society-economy-nature compound ecological system. Based on the foundation of landscape theory, ecological patterns and processes are closely related to the spatial scale. According to previous studies[32,59,60], a grid size of 510 m × 510 m is considered to be the appropriate spatial scale. Eventually, LST and the influencing factors were extracted and calculated at a grid size of 510 m through the fishnet method based on ArcGIS 10.8.

LST retrieval

Following Mhawej and Abunnasr (2022), Daily Ten-ST-GEE was developed using the Google Earth Engine (GEE) to acquire the LST[61]. The main steps involved eliminating clouds, median assessments of the Sentinel-2 and MODIS spectral bands, bandpass adjustments using linear fit regressions, and applying statistical-based robust least squares regression analysis. Field validation showed that the average error was less than 1.0℃. Overall, the development of Daily Ten-ST-GEE using GEE has made it easier and more efficient to obtain accurate LST data, which is an important tool for studying climate change and other environmental issues.

Statistical models

Multiple linear regression (MLR), BRT and GeoD were adopted to reveal the relationships between LST and its influencing factors. Ordinary least squares is a traditional statistical method for quantitatively analyzing the relationship between LST and influencing factors.

The general linear regression model can be expressed as:

$$LST = a + bx + \epsilon \quad (1)$$

where a is constant, b is the regression coefficient, and ϵ is the residual error.

A BRT generates a regression tree algorithm using recursive binary and then boosts to build a large ensemble of small regression trees. Finally, the model captures the nonlinear relationship between LST and the influencing factors[62]. The relative contributions of the influencing factors were determined by the coefficients. The output of partial dependency plots can explain the marginal effects of the influencing factors.

The geographical detector model (GeoD) can determine the interaction effect between two factors[63]. It is established based on spatial heterogeneity. indicates the strength of the influencing factors. It can be expressed as:

$$q = 1 - \frac{1}{N\sigma^2}\sum_{h=1}^{L} N_h \sigma_h^2 \tag{2}$$

where σ^2 is the variance of the response variables; is the number of categories of the response variables; and h=1,2,… is the study population of the response variables.

$q(X1 \cap X2)$ represents the interaction between the influencing factors ($X1$ and $X2$). The interaction mainly includes nonlinear weakening, single-factor nonlinear weakening, nonlinear enhancement, and double-factor enhancement (Table 2).

The development of the BRT model and GeoD model was based on the random forest, dismo, and GD packages of R software. After sensitivity analysis of the BRT model, the learning rate was set to 0.005, and the tree complexity was set to 8.

Results

Patterns of the urban morphology and LST

The patterns of the urban morphology are shown in Fig. 2. Obviously, Chengdu had a monocentric form, whereas Chongqing featured a polycentric form. Specifically, to provide a more quantitative definition of monocentric or polycentric urban forms, Fig. 3 showed the statistical distribution of POP in Chengdu and Chongqing. According to statistical analysis, POP in Chengdu exhibit a unimodal distribution, while in Chongqing exhibit bimodal distribution. This indicated that Chengdu is monocentric, and Chongqing is polycentric. The population density, POI density, and nightlight intensity had high values in the central part of Chengdu. In contrast, Chongqing had some compact main centers. In particular, the population was concentrated in the central part of Chengdu. The area of the highest population density exhibited continuities in Chengdu but displayed small discontinuities in Chongqing. Additionally, the area of highest population density in Chengdu was far beyond that in Chongqing. POI density in Chengdu showed a concentric circle spreading outward with values gradually decreasing, whereas Chongqing had two circles. Regarding the difference in night light intensity, Chengdu exhibited linear aggregation, but central aggregation occurred in Chongqing. From the center part to the edge, the building coverage ratio gradually decreases in Chengdu. The pattern of building height had a opposite distribution in Chengdu. However, the building coverage ratio and building height patterns were similar in Chongqing, high in the commercial center and low in the surrounding areas, as well as the 3D fractal and 3D shape index. The 3D fractal and 3D shape index in Chengdu were higher in the central part and lower in the surrounding areas.

The patterns of summer LST in Chengdu and Chongqing were different (Fig. 4). The summer LST of Chengdu varied from 33.82 °C to 41.68 °C, with an average LST of 37.99 °C. The LST of Chongqing ranged from 31.30 °C to 40.36 °C, with an average LST of 36.67 °C. We also classified these grid sizes into six temperature classes based on the natural break method. The class with the highest LST occupied 17.26% in Chengdu and 9.48% in Chongqing. The sub high class accounted for 21.51% in Chengdu and 23.89% in Chongqing. The area of highest LST was concentrated in the center of Chengdu, whereas the area of highest LST formed a cluster in Chongqing.

Dominant influencing factors

The R^2 values of the multiple linear regression were low for two cities. The variance inflation factor values of the influencing factors were less than 10. The Moran's *I* values for LST exceeded 0.7. This result suggested the high spatial auto-correlation of the LST. Thus, multiple linear regression was weak in explaining the LST.

In terms of the relative contributions, there were similarities and differences between the two cities (Fig. 5). The relative contribution of PIS to LST was 6.5% in Chengdu and 32.2% in Chongqing. The relative contributions of NL to LST in Chengdu and Chongqing were similar (23.8% vs. 27.4%). The contribution of POP to LST in Chengdu (17.6%) was larger than that in Chongqing (9.2%). The contribution of PGS to LST in Chengdu (4.2%) was smaller to that in Chongqing (6.0%). The 3DSI explained 11.9% and 1.7% of the LST variance in Chengdu

Relationship	Interaction
$q(X1 \cap X2) > q(X1) + q(X2)$	Nonlinear enhancement
$q(X1 \cap X2) < Min[q(X1), q(X2)]$	Nonlinear weaken
$q(X1 \cap X2) > Max[q(X1), q(X2)]$	Double-factor enhancement
$Min[q(X1), q(X2)] < q < Max[q(X1), q(X2)]$	Single-factor nonlinear weaken

Table 2. Categories of interaction between two influencing factors.

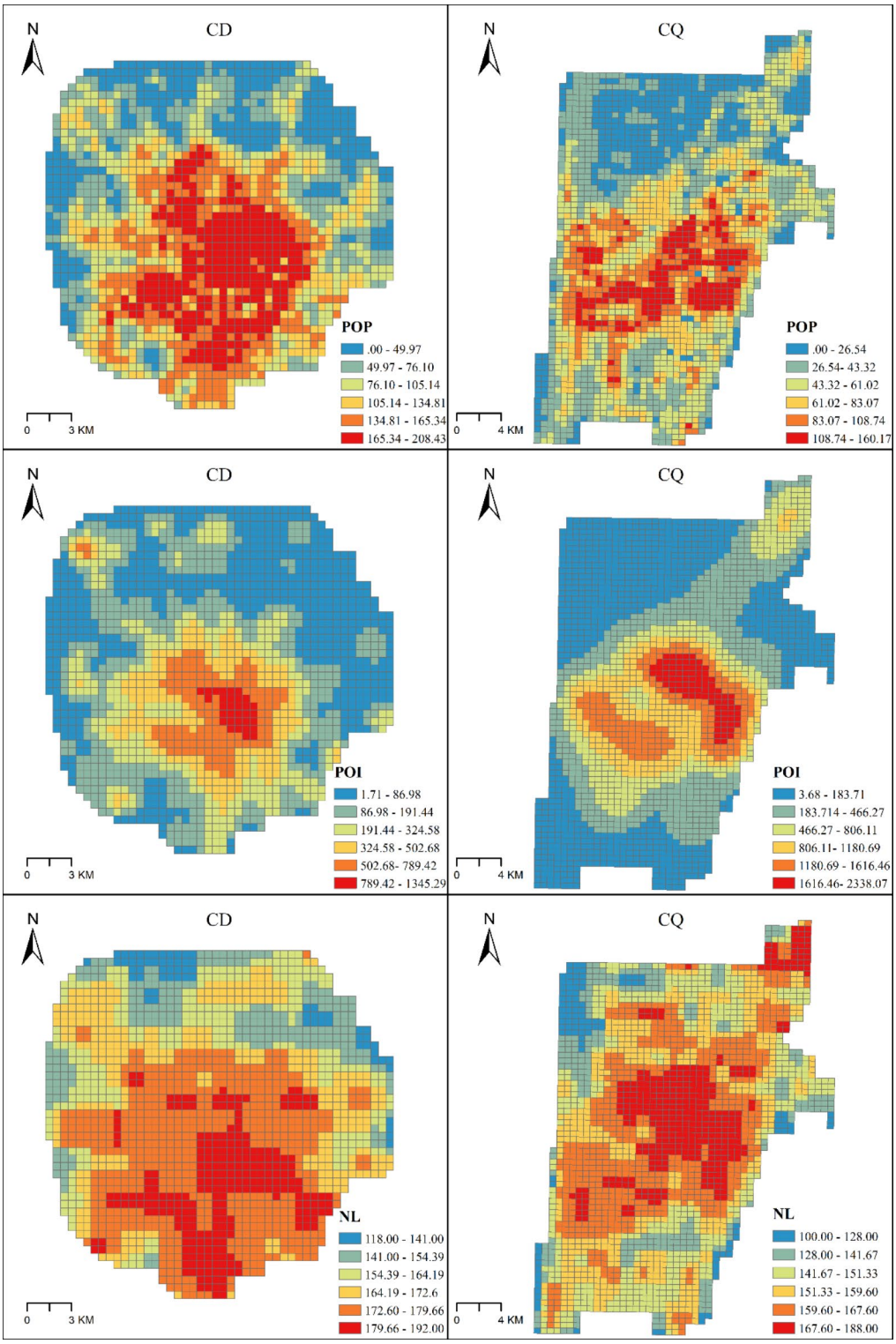


Fig. 2. Spatial distribution of the socio-economic factors.

and Chongqing, respectively. 3DF had the minimum influence on LST, with relative contributions less than 2.0% in the two cities.

As far as the categories, the relative contributions of socio-economic factors (including NL, POP, POI) and topographic factors (USR) to LST were hardly different, while the relative contributions of land use composition (including PIS and PGS) and 3D buildings (including BCR, BH, 3DF and 3DSI) to LST were significantly different in the two cities. The socio-economic factors can explain half of the LST variance in Chengdu and Chongqing. In Chengdu, land use composition explained 10.7% of the LST variance, while it explained 38.2% of

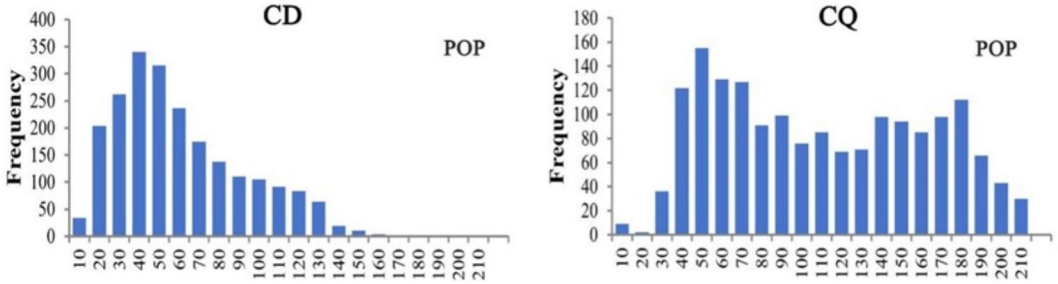


Fig. 3. Distribution of Pop.

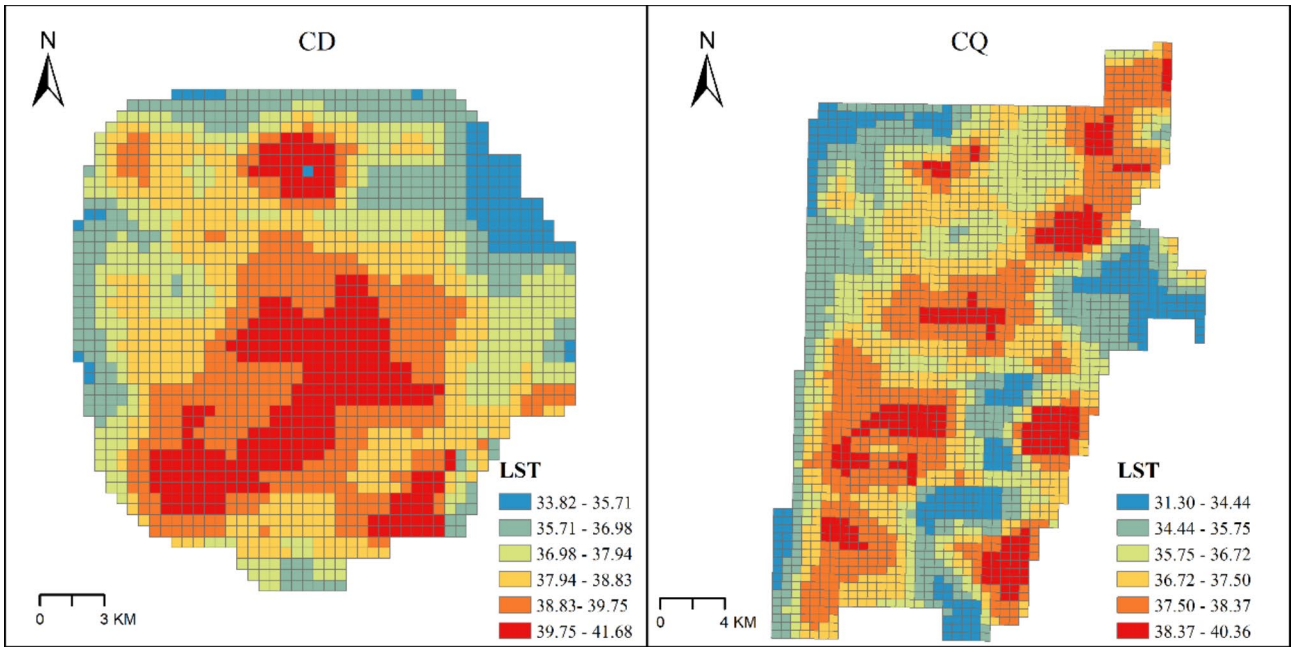


Fig. 4. Distribution of LST.

the LST variance in Chongqing. In contrast, 3D buildings explained 24.8% of the LST variance in Chengdu but only explained 6.2% of the LST variance in Chongqing.

The main contributing factors from the results based on GeoD model were similar to those of the BRT, which indicates that the BRT model is reliable (Fig. 6), especially for Chongqing which has the same top three influencing factors.

Marginal effects of dominant influencing factors

The marginal effects between these influencing factors (including NL, POP, USR, and PIS) and LST were similar but differed in magnitude and range of action (Fig. 7). In Chengdu, the LST slowly increased when the NL ranged from 120 to 175 and then sharply decreased and slightly increased when the NL exceeded 180. However, the LST gradually increased when NL varied from 100 to 180 and then remained stable in Chongqing. Similar to POP, the LST tended to decrease as POP increased in the two cities. The action scopes of POP were 100–150 in Chengdu and 0-150 in Chongqing. As the PIS increased from 0 to 100, the LST continuously increased.

The relationships between LST and POI and 3DSI in the two cities were different. Specifically, the relationship between POIs and LST was complex in Chengdu. When the POI was less than 100, the relationship was positive, but it became negative when the POI exceeded 100. POI showed a positive correlation with the LST in Chongqing. In Chengdu, a sharp increase in the LST was observed when the 3DSI ranged from 2.6 to 4.2 and LST remained stable when the 3DSI exceeded 4.2. However, the LST was not sensitive to changes in the 3DSI in Chongqing.

The LST regulation amplitudes for the NL and PIS in Chengdu were less than those in Chongqing. The LST regulation amplitudes for the NL/PIS were 1.0 °C/0.4 °C and 2.1 °C/2.0 °C in Chengdu and Chongqing, respectively. The LST regulation amplitude for the POP in Chengdu (2.0 °C) were larger than that in Chongqing (1.0 °C). Additionally, the LST regulation amplitude for the 3DSI reached 1.1 °C in Chengdu.

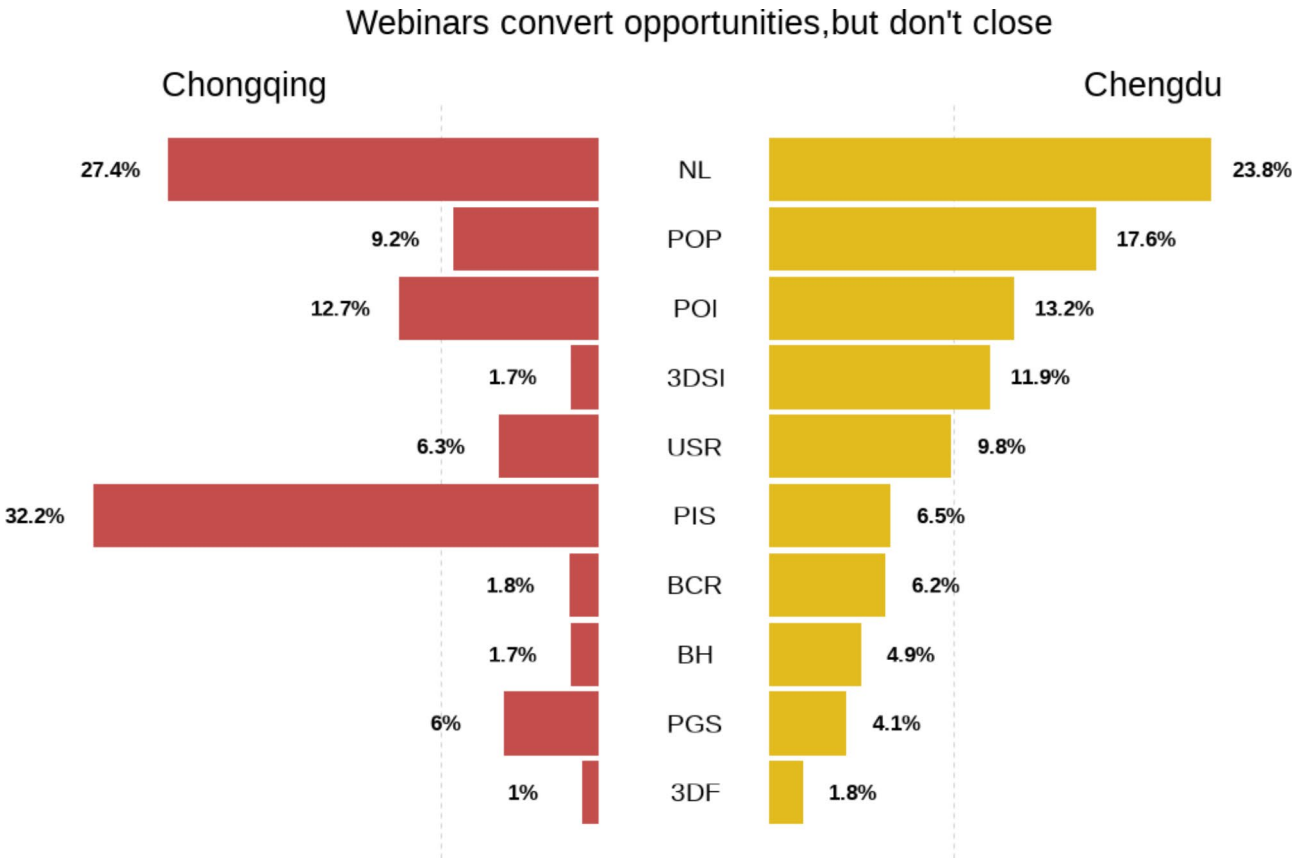


Fig. 5. Contributions (%) of the various indicators to the LST based on BRT.

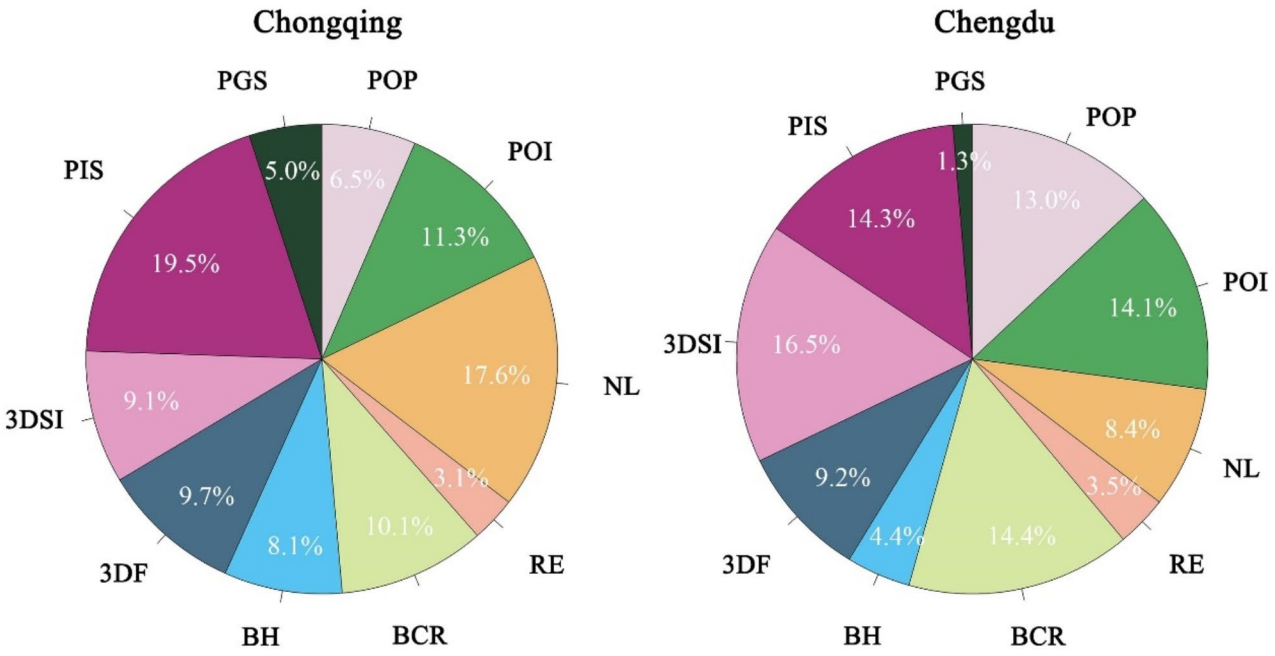


Fig. 6. Contributions (%) of the various indicators to the LST based on GeoD model.

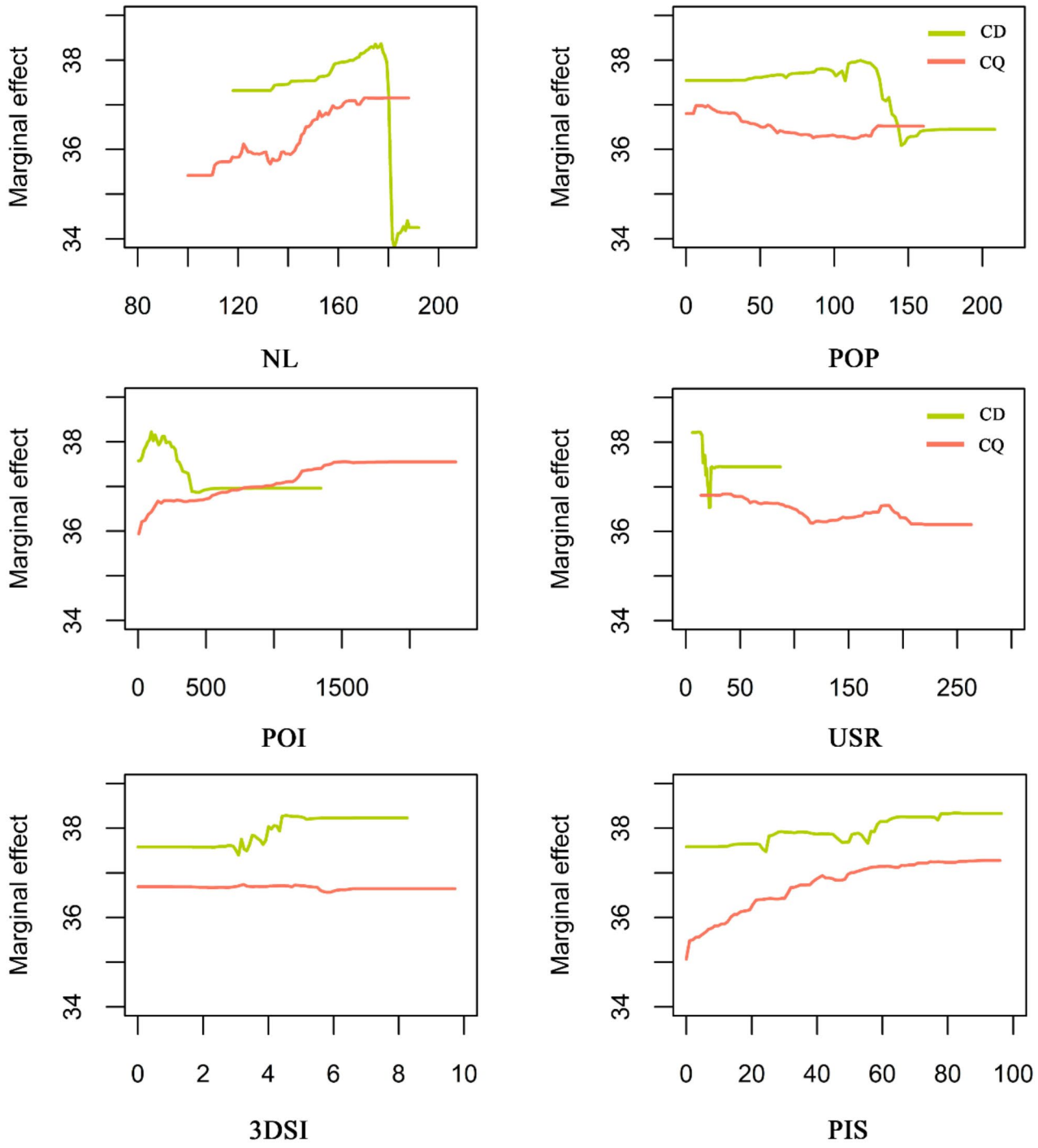


Fig. 7. Marginal effects of the most important explanatory variables in Chengdu and Chongqing.

Interactive effects between the influencing factors

For the NL and PIS, the interaction effect was nonlinear weaken (NW) in Chengdu and Chongqing, suggesting that LST would be decreased when the NL included PIS (Fig. 8). In terms of socio-economic variables, POP and POI, POI and NL both exhibited nonlinear weaken in Chengdu and Chongqing. POP and NL were double-factor enhanced in Chengdu but nonlinearly weakened in Chongqing. In terms of building-related variables, the interaction effect of the BCR and BH showed a nonlinear weaken, which suggested that LST would be decreased when the BCR included BH in Chengdu and Chongqing. Similarly, the 3DF and 3DSI experienced nonlinear weaken.

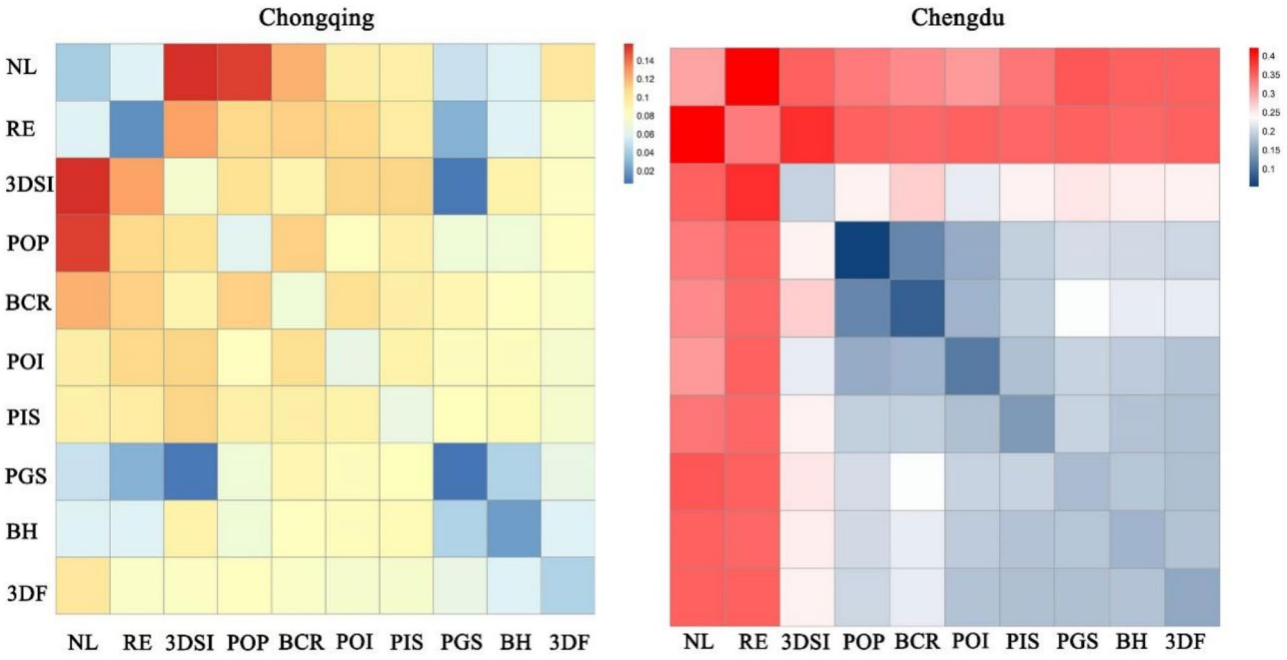


Fig. 8. Interactive effects of the explanatory variables in Chengdu and Chongqing.

Discussion

Influence of urban morphology indicators on LST

Terrain had a distinct impact on urban development patterns. Many plain cities tend to exhibit a monocentric development model, whereas mountainous cities often display a polycentric development model. Chengdu, for example, has maintained a monocentric urban pattern with urban expansion outward due to its flat terrain[47]. Chongqing chose a polycentric urban pattern due to its complex terrain, limited land space, etc[64]. The monocentric urban pattern causes excessive population to be concentrated in the core. Similar distribution applied to buildings, impervious surfaces, and restaurants. However, the polycentric urban pattern was multipoint concentration. Chongqing had several strong subcenters, and the population was characterized by site-specific concentration and overall dispersion. This is consistent with the other urban morphology indicators.

The differences in urban development patterns lead to spatial differences in LST. Distributions of LST correspond to the pattern of each city. Socio-economic factors, such as NL and POI, had a negative impact on LST, with similar relative importance but different magnitudes and ranges of action. NL and POI represent human activity intensity, and high values in NL and POI indicate more anthropogenic heat emissions. Ultimately, more anthropogenic heat emissions lead to a rise in LST. According to previous studies, the anthropogenic heat flux in Chengdu decreased from the urban core to the suburbs, whereas it was concentrated in several centers in Chongqing[64]. POP showed a negative correlation with LST, which is consistent with previous studies[47]. UHIs are largely explained by climate and population at the global scale[65]. At the city scale, numerous studies have found that population density was positively correlated with LST[43], but the relationship was less significant in the United States[31]. Another possible reason was that a higher population density implies a high building height and floor area ratio. A study conducted on a plain city found that high-rise buildings decrease the LST due to casting more shadows[66].

Generally, the higher the BCR is, the higher the LST is[60]. In contrast, the area with high-rise buildings tend to have lower LST. Building morphology factors have less explanation for LST in Chongqing compared to that in Chengdu. A possible explanation was that complex terrain has counteractive effects on other influencing factors and thereby decreases the contribution of building morphology in Chongqing. Indeed, the contribution of USR exceeded the sum contribution of building morphology factors (BCR, BH, 3DSI, and 3DF). We found that the 3DSI was monotonically positive with LST. Previous studies have proven that the rise of shape coefficient, a similar indicator to 3DSI, increases the LST due to the increased surface area for conducting heat[40].

Land cover composition in Chongqing, such as PIS and PGS, was superior to that in Chengdu. These differences can be explained from the following aspects. First, there is a higher proportion of impervious surfaces in Chengdu than that in Chongqing due to the lower cost of urban development in Chengdu. A high proportion of impervious surfaces leads to a higher LST, since the impervious surface has a lower albedo and a high solar irradiation absorption[67]. Second, the building aggregation form has an adverse impact on LST for two cities. In Chengdu, low rise and high density buildings agglomerated in the core area. This pattern means a high building coverage ratio, which increases the LST in Chengdu. In contrast, high-rise and high height buildings were dispersed in Chongqing. High height buildings coupled with complex terrain reduce ground solar radiation and cast more shaded areas.

Implications for urban planning

This study provides insights into how to decrease the LST in cities by optimizing urban morphology. For plain cities, we suggest relieving a portion of the population to the suburbs. By relieving a portion of the population to the suburbs, it not only reduces the population under high temperature stress but also reduces anthropogenic heat emission in the urban central areas. Additionally, continually optimizing the green space layout and reducing PIS can decrease LST. Optimizing the green space layout could also provide shade and cooling effects. For mountainous cities, buildings with large volumes are common. High-rise buildings with a large surface area and low albedo absorb more heat than low-rise buildings, which contributes to the LST in these cities. Thus, decreasing the 3DSI would be an effective way to decrease the LST.

Limitations and future research

More urban energy balance observation experiments are needed to explore the change mechanism of the LST. Previous studies suggested that local climate affected the energy balance processes and therefore modified the LST. In the future, air temperature, humidity and wind should be considered. Furthermore, how the relationships between urban morphology and LST varied with different local climates is worth of further exploration. Additionally, this case study focused on specific region and time, which might limit the generalization of the results. Future research can be conducted on a larger scale of time and space.

Conclusions

This study explored the influences of comprehensive urban forms on summer LST by comparative analysis in Chengdu and Chongqing. The results showed that high LST was concentrated in the central part of Chengdu, while multicenter aggregation of high LST occurred in Chongqing. Socio-economic factors had the largest contributions to LST in both cities, with relative contributions of approximately 50%. The land use composition significantly influenced the LST in Chengdu, while it had less impact on the LST in Chongqing. Urban building morphology contributed greatly to the LST in Chongqing but had less effect on the LST in Chengdu. The BRT analysis results found that the socio-economic factors and terrain had similar contributions for Chengdu and Chongqing, but the contributions of land use composition and building morphology varied greatly for two cities. Different UHI mitigation strategies should be implemented for two cities, such as optimizing the green space layout and reducing PIS in Chengdu and strengthening 3D building management in Chongqing.

Data availability

The data that support the findings of this study are available on request from the corresponding author, upon reasonable request.

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Declarations

Competing interests
The authors declare no competing interests.

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